

Foundational Mathematics

From counting to change

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Chapter 1 - Arithmetic

fundamental ideas

Arithmetic is the language of calculation.

Definition

Arithmetic is the branch of mathematics concerned with the basic operations on numbers: addition, subtraction, multiplication, and division. In a broader sense, it also includes exponentiation, roots, and logarithms. Put simply, arithmetic is the part of mathematics that studies how numbers behave under calculation.

Core operations

Arithmetic deals with how numbers can be combined, transformed, and compared.

The most important operations are addition, subtraction, multiplication, and division. These operations let us describe quantities, compare values, and model change. From these basic rules, more advanced ideas in algebra and calculus are built.

Examples

A few familiar examples are:

- $7 + 5 = 12$
- $9 - 4 = 5$
- $3 \times 6 = 18$
- $20 \div 4 = 5$

Notice that each statement gives us a different kind of relationship: combining, removing, scaling, and distributing. Arithmetic is not merely about computing, but about understanding how operations transform quantities.

Number Theory Foundations

Beyond basic calculation, arithmetic leads into Number Theory.

Two fundamental concepts are:

- Prime Factorization: Every integer greater than 1 is either a prime number or can be represented as a unique product of prime numbers.

For example, $60 = 2^2 \times 3 \times 5$.

- GCD and LCM: The Greatest Common Divisor (GCD) is the largest positive integer that divides each of the integers. The Least Common Multiple (LCM) is the smallest positive integer that is divisible by both.

Modular Arithmetic

Often called "clock math," modular arithmetic deals with integers that "wrap around" upon reaching a certain value, called the modulus.

For example, in modulo 12 (like a clock), $10 + 5 \equiv 3 \pmod{12}$.

This system is the cornerstone of modern cryptography and computer science.

Why arithmetic matters

Arithmetic is often the first structured way a person learns to reason with numbers. It teaches consistency, precision, and symbolic thinking. Without it, topics such as algebra, geometry, probability, and calculus would not have the same foundation.

Types of arithmetic

- a) Integer arithmetic
- b) Rational number arithmetic
- c) Real number arithmetic
- d) Approximation and estimation

Different number systems lead to different rules and subtleties. For example, operations on whole numbers are not always identical to those on fractions or irrational numbers, even though the basic ideas remain similar.

Properties of arithmetic

Arithmetic is governed by a few basic properties that make calculations consistent. These include the commutative property, which says that order does not matter in addition or multiplication, and the associative property, which says that grouping does not matter.

There is also the distributive property, which connects multiplication with addition:

$$a(b + c) = ab + ac$$

These properties are essential because they explain why arithmetic behaves the way it does and why algebraic manipulation is possible.

Worked example

Suppose we want to simplify $6 \times (4 + 3)$. Using the distributive property, we can write:

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3 = 24 + 18 = 42$$

This shows how arithmetic can be reorganized without changing the value. Such transformations are central to algebra and problem solving.

Key idea

Everything else in mathematics often grows from the simple operations of arithmetic. Even when notation becomes more advanced, the underlying idea is the same: numbers can be transformed in consistent, meaningful ways. This is why arithmetic remains one of the most foundational parts of mathematical thinking.

Summary

Arithmetic is the study of the basic operations on numbers and the rules that govern them. It gives us a language for calculation, a foundation for algebra, and the structure needed for deeper mathematical reasoning.

Next topics

- Number system
- Limits and continuity
- Differential equations

Chapter 2 - Number System

foundations

The number system gives structure to all calculation.

The Number Hierarchy

Mathematics organizes numbers into sets, where each set typically expands upon the previous one. This hierarchy is represented as:

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$

Natural Numbers $\mathbb{N}$

The natural numbers are the counting numbers: $\{1, 2, 3, 4, \dots\}$.

They are the simplest numbers and are used for counting discrete objects.

Integers $\mathbb{Z}$

Integers include positive natural numbers, their negatives, and zero:

$\{\dots, -2, -1, 0, 1, 2, \dots\}$. They allow for the representation of direction and magnitude (e.g., debt or temperature).

Rational Numbers $\mathbb{Q}$

Rational numbers can be expressed as a fraction $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$. Examples include $\frac{1}{2}$, $-\frac{5}{7}$, and 0.75 (which is $\frac{3}{4}$).

Irrational Numbers

Irrational numbers cannot be expressed as a fraction of two integers. Their decimal expansions are non-repeating and infinite.

Famous examples include $\sqrt{2}$ and $\pi \approx 3.14159$.

Real Numbers $\mathbb{R}$

Real numbers are the union of rational and irrational numbers. They correspond to every single point on a continuous number line.

Complex Numbers $\mathbb{C}$

Complex numbers extend the real numbers by introducing the imaginary unit i , defined by the property $i^2 = -1$. A complex number is written in the form $a + bi$, where a and b are real numbers.

Key idea

The number system is not just a list of categories. It shows how mathematical objects can be organized by structure, and how different kinds of numbers respond to the operations we perform on them.

Chapter 3 - Limits and Continuity

calculus foundations

Limits ask what happens as a value approaches a point.

Definition

The limit of a function describes the value a function approaches as its input gets closer and closer to some point. It is a central idea in calculus because it captures behavior near a point without requiring exact value at that point.

Example

Consider the function $f(x) = \frac{x^2 - 1}{x - 1}$. For $x \neq 1$, this simplifies to $x + 1$. As $x \rightarrow 1$, the function approaches 2, even though the original expression is undefined at $x = 1$.

Continuity

A function is continuous at a point if the limit exists, the function value exists, and they are equal. Intuitively, a continuous function can be drawn without lifting the pen.

Why it matters

Limits and continuity are the foundation of derivatives and integrals. They allow us to study motion, rates of change, and accumulation in a precise mathematical way.

Key idea

A limit is about approach, not necessarily equality at the point itself. This idea is what gives calculus its power: it allows us to understand behavior near values that may not be directly defined there.

Chapter 4 - Differential Equations

applied calculus

Differential equations express how quantities change.

Definition

A differential equation is an equation involving a function and one or more of its derivatives. Such equations describe how a quantity changes in relation to itself or to other variables.

Why they matter

Differential equations appear in physics, biology, economics, and engineering. They can be used to model processes such as population growth, cooling, motion, and electrical circuits.

Simple example

The equation $dy/dx = ky$ describes exponential growth or decay. If $k > 0$, the quantity grows; if k

Types

There are many kinds of differential equations, including first-order equations, second-order equations, linear equations, and separable equations. Different types often require different techniques for solving them.

Key idea

Differential equations are not just symbolic puzzles. They express change in a language that can model real systems. The goal is often not only to solve them, but to interpret what the solution means in context.

Afterword - Why Mathematics?

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